

Artificial Intelligence in Pediatric Radiology

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CHLA Radiology Grand Rounds

October 8, 2026

Objectives

- Describe the big players in radiology AI, both general and pediatric
- Summarize what radiology AI does well and what remains unresolved

Methods

Journals, Preprints: PubMed, Embase, arXiv, medRxiv

Conferences: NeurIPS, ICLR, ICML, CVPR, MICCAI, MIDL

Datasets: Hand-curated list

Newsletters: The Imaging Wire, RSNA News, TLDR AI, TLDR Tech, Signify Research, ESR / ECR, Radiology Business

Commercial: FDA list of AI-enabled devices

Date range: January 1 2008 – September 30 2026

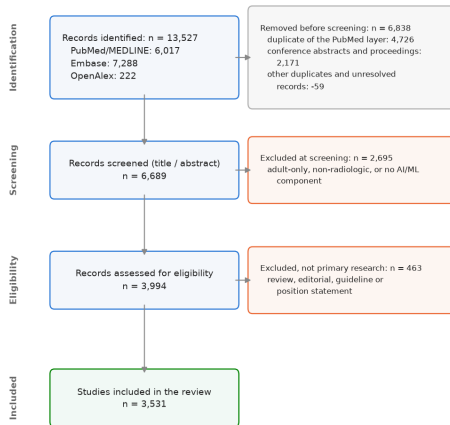
Search strategy: radiology AND AI [AND pediatric]

radiology: (radiology[tiab] OR radiological[tiab] OR radiograph*[tiab] OR "medical imaging"[tiab] OR "diagnostic imaging"[tiab] OR tomography[tiab] OR "magnetic resonance"[tiab] OR MRI[tiab] OR "MR imaging"[tiab] OR "MR images"[tiab] OR MRIs[tiab] OR "MR image"[tiab] OR "MR scan"[tiab] OR "MR scans"[tiab] OR "computed tomography"[tiab] OR CT[tiab] OR ultrasound[tiab] OR ultrasonograph*[tiab] OR sonograph*[tiab] OR echocardiograph*[tiab] OR mammograph*[tiab] OR "chest x-ray"[tiab] OR "x-ray"[tiab] OR fluoroscop*[tiab] OR scintigraph*[tiab] OR "positron emission"[tiab] OR PET[tiab] OR SPECT[tiab] OR angiograph*[tiab] OR neuroimaging[tiab] OR tractograph*[tiab] OR elastograph*[tiab] OR "fMRI"[tiab] OR "functional MRI"[tiab] OR "functional magnetic resonance"[tiab] OR connectome*[tiab] OR "diffusion tensor"[tiab] OR DTI[tiab] OR "bone age"[tiab] OR "skeletal age"[tiab] OR "skeletal maturity"[tiab] OR "barium enema"[tiab])

AI: ("artificial intelligence"[tiab] OR "machine learning"[tiab] OR "deep learning"[tiab] OR "convolutional neural network"[tiab] OR "convolutional neural networks"[tiab] OR "neural network"[tiab] OR "neural networks"[tiab] OR "computer-aided diagnosis"[tiab] OR "computer aided diagnosis"[tiab] OR "computer-aided detection"[tiab] OR radiomic*[tiab] OR "computer vision"[tiab] OR "large language model"[tiab] OR "large language models"[tiab] OR "foundation model"[tiab] OR "foundation models"[tiab] OR "transfer learning"[tiab] OR "vision transformer"[tiab] OR "self-supervised"[tiab] OR "random forest"[tiab] OR "support vector machine"[tiab] OR "gradient boosting"[tiab] OR "natural language processing"[tiab])

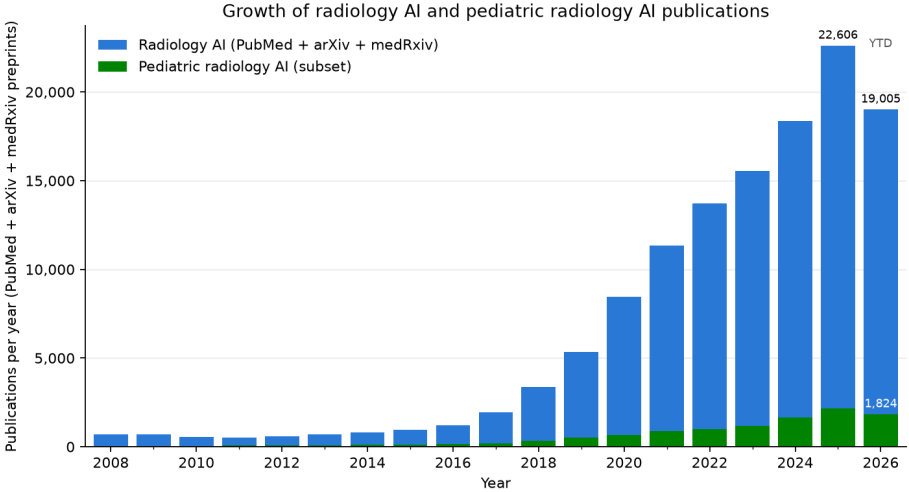
pediatric: (pediatric*[tiab] OR paediatric*[tiab] OR child*[tiab] OR infant*[tiab] OR neonat*[tiab] OR adolescen*[tiab] OR fetal[tiab] OR foetal[tiab] OR fetus*[tiab] OR foetus*[tiab] OR newborn*[tiab] OR preterm[tiab] OR "premature infant"[tiab] OR "premature infants"[tiab] OR "premature birth"[tiab] OR "premature neonate"[tiab] OR "premature neonates"[tiab] OR prenatal[tiab] OR antenatal[tiab] OR perinatal[tiab] OR youth[tiab] OR juvenile[tiab] OR "children's hospital"[tiab] OR schoolchild*[tiab] OR toddler*[tiab] OR baby[tiab] OR babies[tiab])

Study selection

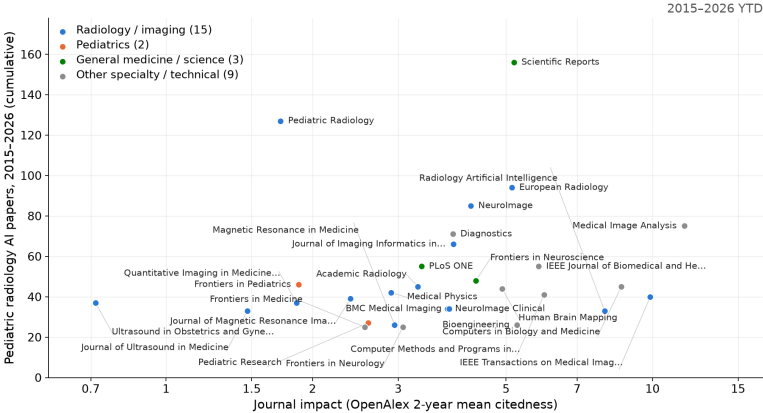


PubMed and Embase layers complete (2005-2026). Conference material is retrieved and reported but excluded by the publication-form criterion.

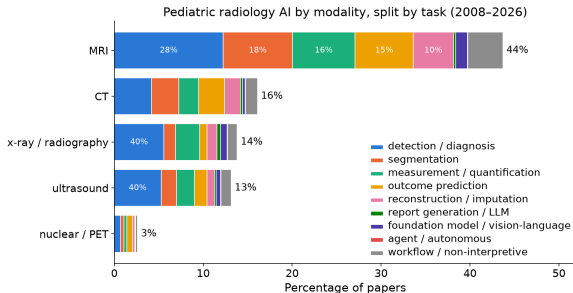
Radiology AI publication counts



Which journals publish pediatric radiology AI



Pediatric radiology AI publications, stratified by modality and task

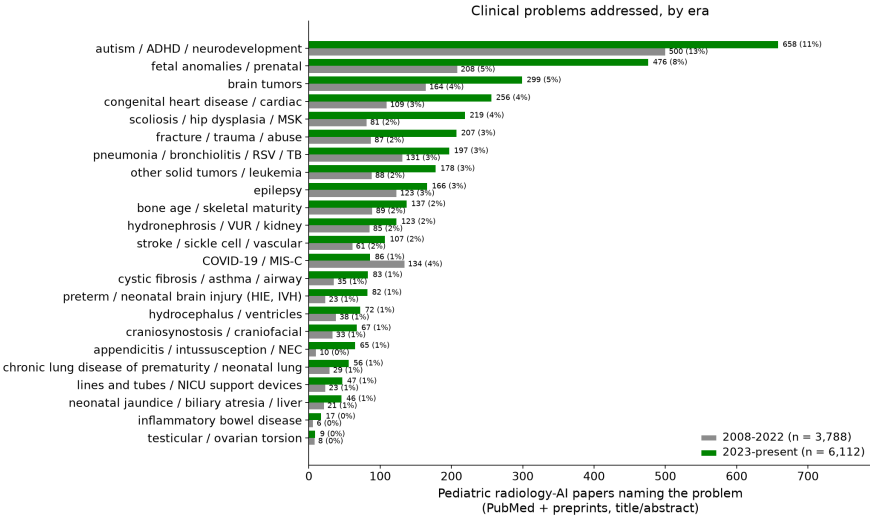


- **detection / diagnosis:** predict discrete disease labels (e.g., fracture, pneumonia, tumor, triage)
- **segmentation:** outline a structure or lesion
- **measurement / quantification:** measure a number from the image (e.g., bone age, Cobb angle, organ volume, fetal biometry)

- **outcome prediction:** a future risk, response, or survival estimate from the image (prognosis, radiomics signatures)
- **reconstruction / imputation:** generate a better or missing image
- **report generation / LLM:** text: draft, summarize, or extract from radiology reports

- **foundation model / vision-language:** a large pretrained model reused across tasks
- **agent / autonomous:** multi-step actions taken without a human in the loop
- **workflow / non-interpretive:** protocoling, scheduling, ordering, decision support, education

Pediatric radiology AI publications, stratified by medical problem



Most-cited radiology AI papers, 2008-2022

Cites	FWCI	Paper	Venue	Clinical problem	What it answers
5,797	443.4	Swin-Unet: Unet-like Pure Transformer for Medical Image Segmentation (2021)	ECCV Workshops	general (multi-organ CT, cardiac MRI)	<i>segmentation</i> : Can a pure transformer, with no convolutions, outline abdominal organs on CT and heart chambers on MRI as well as a U-Net?
2,628	288.8	Swin UNETR: Swin Transformers for Semantic Segmentation of Brain Tumors in MRI Images (2022)	BrainLes@MICCAI	brain tumors (glioma)	<i>segmentation</i> : Can a transformer encoder outline glioma subregions (enhancing tumor, edema, necrotic core) on multiparametric brain MRI? (BraTS 2021)
2,782	262.1	COVID-Net: a tailored deep convolutional neural network design for detection of COVID-19 cases from chest X-ray images (2020)	Scientific Reports	COVID-19 / pneumonia	<i>diagnosis</i> : Can a purpose-built CNN tell COVID-19 pneumonia from other pneumonia and normal chest X-rays? (COVID-Net, open source)
3,710	230.9	ChestX-Ray8: Hospital-Scale Chest X-Ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases (2017)	Computer Vision and Pattern Recognition	thoracic disease on chest X-ray	<i>diagnosis</i> : Can labels mined automatically from radiology reports train a model to classify and localize eight common thoracic findings? (NIH ChestX-ray dataset, 108,948 images)
10,389	219.2	nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation (2020)	Nature Methods	general (any organ or lesion)	<i>segmentation</i> : Can one self-configuring method segment any organ or lesion without hand tuning? (nnU-Net)
2,036	206.6	Covid-19: automatic detection from X-ray images utilizing transfer learning with convolutional neural networks (2020)	Physical and Engineering Sciences in Medicine	COVID-19 / pneumonia	<i>diagnosis</i> : Can CNNs pretrained on everyday photographs detect COVID-19 on chest X-ray from only a few hundred positive cases?
5,662	156.0	Computational Radiomics System to Decode the Radiographic Phenotype (2017)	Cancer Research	cancer (any site)	<i>radiomics</i> : Can quantitative image features (shape, intensity, texture) be extracted in a standardized, reproducible way across sites? (PyRadiomics, open source)
2,145	135.7	MultiResUNet : Rethinking the U-Net Architecture for Multimodal Biomedical Image Segmentation (2019)	Neural Networks	general (multiple image types)	<i>segmentation</i> : Does a redesigned U-Net segment more reliably across five biomedical image types, including brain MRI?
2,126	119.7	CE-Net: Context Encoder Network for 2D Medical Image Segmentation (2019)	IEEE Transactions on Medical Imaging	general (2D images: lung CT, retina)	<i>segmentation</i> : Does adding a context-encoder module to a U-Net improve 2D segmentation of structures such as the lungs on CT and the optic disc and vessels on retinal images?
9,731	86.7	UNet++: A Nested U-Net Architecture for Medical Image Segmentation (2018)	DLMIA/ML-CDS@MICCAI	general (nodule, liver, polyp, nuclei)	<i>segmentation</i> : Do nested, redesigned skip connections make U-Net more accurate on lung nodules, liver, colon polyps and cell nuclei?

FWCI = field-weighted citation impact (OpenAlex): citations relative to the world average for papers of the same field and year (1 = average)

Most-cited radiology AI papers, 2026 (year to date)

Cites	FWCI	Paper	Venue	Clinical problem	What it answers
59	111.4	A generalizable foundation model for analysis of human brain MRI	Nature Neuroscience	brain (multiple disorders)	<i>foundation model</i> : Can one model pretrained on large unlabeled brain MRI collections be adapted to many downstream brain tasks with little labeled data?
12	61.4	Explainability-Aligned Reliability-Weighted Fuzzy Ensemble for Automated Cervical Cancer Classification	International Journal of Intelligent Systems	cervical cancer	<i>diagnosis</i> : Can a reliability-weighted ensemble of CNNs classify cervical cancer images and show which image regions drove each call?
21	58.8	A hybrid deep learning framework based on VGG19 and U-Net for accurate brain tumor segmentation in MRI images	Magnetic Resonance Letters	brain tumors	<i>segmentation</i> : Does pairing a pretrained VGG19 encoder with a U-Net decoder give more accurate brain tumor outlines on MRI?
14	50.1	MedVersa: A Generalist Foundation Model for Diverse Medical Imaging Tasks	NEJM AI	general (multiple modalities)	<i>foundation model</i> : Can a single generalist model handle many imaging tasks (report drafting, detection, segmentation) across chest X-ray, CT, MRI and dermatology images?
15	49.5	Explainable AI-driven MRI-based brain tumor classification: a novel deep learning approach	Frontiers Artif. Intell.	brain tumors	<i>diagnosis</i> : Can a CNN classify brain tumor type on MRI and highlight the image regions behind each prediction?
13	46.0	A deep learning framework for breast cancer diagnosis using Swin Transformer and Dual-Attention Multi-scale Fusion Network	Scientific Reports	breast cancer	<i>diagnosis</i> : Can a Swin Transformer with multi-scale attention fusion separate benign from malignant breast lesions on imaging?
13	44.2	Multi-class eye disease classification using deep learning EfficientNetB0 fusion techniques	Scientific Reports	eye disease (retinal images)	<i>diagnosis</i> : Can fused EfficientNet models classify several eye diseases from retinal images? (Ophthalmic, outside radiology proper)
12	39.6	3D-UNet+: An enhanced 3D U-Net framework for brain tumor segmentation with systematic hyperparameter tuning	Neural computing & applications (Print)	brain tumors	<i>segmentation</i> : Does systematic hyperparameter tuning of a 3D U-Net improve brain tumor segmentation on multiparametric MRI?
13	30.9	Deep Learning Approaches for Brain Tumor Classification in MRI Scans: An Analysis of Model Interpretability	Applied Sciences	brain tumors	<i>diagnosis</i> : Which CNN brain tumor classifiers on MRI give explanations that point at the tumor rather than at irrelevant parts of the image?
19	27.1	BIBSNet: A deep learning baby image brain segmentation network for MRI scans	Developmental Cognitive Neuroscience	neurodevelopment (infant brain)	<i>segmentation</i> : Can one network segment infant brain MRI (gray and white matter, deep structures) across the first years of life, when tissue contrast is still changing? (BIBSNet, open source)

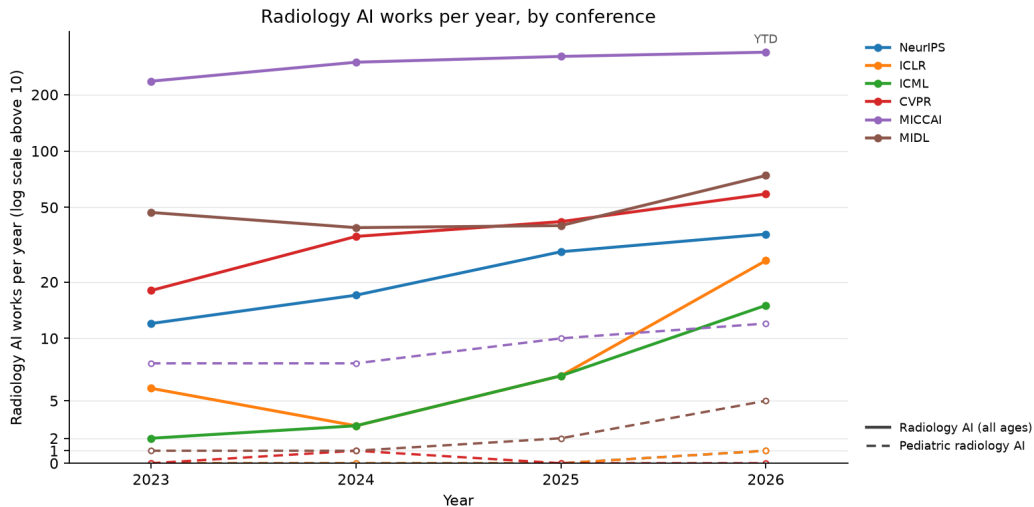
Most-cited pediatric radiology AI papers, 2008-2022

Cites	FWCI	Paper	Venue	Clinical problem	What it answers
426	310.7	Performance of a Deep-Learning Neural Network Model in Assessing Skeletal Maturity on Pediatric Hand Radiographs. (2017)	Radiology	bone age / skeletal maturity	<i>measurement</i> : Can a CNN read bone age from a hand radiograph as accurately as pediatric radiologists using the Greulich–Pyle atlas?
432	305.5	Fully Automated Deep Learning System for Bone Age Assessment (2017)	Journal of digital imaging	bone age / skeletal maturity	<i>measurement</i> : Can a fully automated pipeline (find the hand, standardize the image, read it) assign bone age with no manual steps, fast enough for clinical use?
407	206.8	Deep learning for automated skeletal bone age assessment in X-ray images (2017)	Medical Image Anal.	bone age / skeletal maturity	<i>measurement</i> : Can a deep network estimate bone age directly from radiographs, without hand-crafted features, across sexes and ethnicities? (BoNet)
144	69.0	Computerized Bone Age Estimation Using Deep Learning Based Program: Evaluation of the Accuracy and Efficiency. (2017)	AJR. American journal of roentgenology	bone age / skeletal maturity	<i>measurement</i> : Is a commercial deep-learning bone-age program (VUNO) accurate, and does it shorten radiologist reading time?
353	55.0	Standard Plane Localization in Fetal Ultrasound via Domain Transferred Deep Neural Networks (2015)	IEEE journal of biomedical and health informatics	fetal anomalies / prenatal	<i>detection</i> : Can a CNN find the frame showing the fetal abdominal standard plane (where biometry is measured) in an ultrasound video, using layers transferred from a photo-trained network to cope with little data?
196	54.4	Automatic Fetal Ultrasound Standard Plane Recognition Based on Deep Learning and IoT (2021)	IEEE Transactions on Industrial Informatics	fetal anomalies / prenatal	<i>detection</i> : Can deep learning recognize the standard fetal ultrasound views automatically, reducing dependence on sonographer experience?
449	35.4	A transfer learning method with deep residual network for pediatric pneumonia diagnosis (2020)	Comput. Methods Programs Biomed.	pneumonia / bronchiolitis / RSV / TB	<i>diagnosis</i> : Can a residual CNN with transfer learning diagnose pneumonia on pediatric chest X-rays (children aged 1–5, Guangzhou dataset)?
236	28.9	Evaluation of deep convolutional neural networks for automatic classification of common maternal fetal ultrasound planes (2020)	Scientific Reports	fetal anomalies / prenatal	<i>detection</i> : Can CNNs sort routine maternal–fetal ultrasound images into the standard planes (fetal brain, abdomen, femur, thorax, cervix) as well as a sonographer? (public FETAL_PLANES_DB)
460	26.4	The RSNA Pediatric Bone Age Machine Learning Challenge. (2019)	Radiology	bone age / skeletal maturity	<i>bone age</i> : How well can crowdsourced AI estimate bone age? (RSNA challenge, 260 teams)
258	25.8	An ensemble of neural networks provides expert-level prenatal detection of complex congenital heart disease (2021)	Nature Medicine	congenital heart disease / cardiac	<i>detection</i> : Can an ensemble of networks flag complex congenital heart disease on routine fetal screening ultrasound views as well as fetal cardiologists?

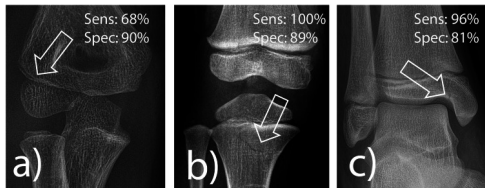
Most-cited pediatric radiology AI papers, 2026 (year to date)

Cites	FWCI	Paper	Venue	Clinical problem	What it answers
3	45.1	Wavelet-Based Frequency Replacement and Edge Enhancement for Semi-Supervised Fetal Ultrasound Image Segmentation	Journal of ultrasound in medicine	fetal anomalies / prenatal	<i>segmentation</i> : Can fetal head and pubic symphysis be segmented on ultrasound from only a few labeled images, using frequency and edge cues to learn from unlabeled ones?
19	27.1	BIBSNet: A deep learning baby image brain segmentation network for MRI scans	Developmental Cognitive Neuroscience	neurodevelopment (infant brain)	<i>segmentation</i> : Can one network segment infant brain MRI (gray and white matter, deep structures) across the first years of life, when tissue contrast is still changing? (BIBSNet, open source)
4	27.1	Beyond Benchmarks of IUGC: Rethinking Requirements of Deep Learning Methods for Intrapartum Ultrasound Biometry from Fetal Ultrasound Videos	Medical Image Anal.	labor / intrapartum (prenatal)	<i>measurement</i> : What do deep-learning methods need to measure labor progress automatically from intrapartum ultrasound video where trained sonographers are scarce? (IUGC challenge, 774 videos)
3	26.2	Zero-shot performance of a general-purpose vision-language model for pediatric appendicitis diagnosis.	Pediatric Radiology	appendicitis / intussusception / NEC	<i>vision-language</i> : Can a general-purpose vision-language model (Llama 4), with no task-specific training, diagnose pediatric appendicitis from ultrasound images, report text and clinical data? (294 patients)
5	24.1	Deep learning assessment of fetal brain maturation on 3D ultrasound volumes in early-onset fetal growth restriction	Ultrasound in Obstetrics and Gynecology	fetal anomalies / prenatal	<i>measurement</i> : Can deep learning grade fetal brain maturation on 3D ultrasound in early-onset growth restriction, and does a maturation delay predict neonatal complications?
3	18.1	A multi-task deep learning and radiomics framework for fetal anatomical structure detection and classification in ultrasound imaging	Scientific Reports	fetal anomalies / prenatal	<i>detection</i> : Can one model locate and label nine fetal brain and craniofacial structures on first-trimester ultrasound to support early anomaly screening? (4,532 scans, 9 centers)
2	18.1	Deep learning-based cerebellar segmentation on T2-weighted fluid-attenuated inversion recovery magnetic resonance imaging for detecting cerebellar hypoplasia/atrophy in infants	Pediatric Radiology	neurodevelopment (cerebellar hypoplasia / atrophy)	<i>segmentation</i> : Can automated cerebellar volume on infant T2-FLAIR MRI flag cerebellar hypoplasia or atrophy against age-expected size?
3	14.6	Deep Learning-Based Multi-Class Pediatric Wrist Fracture Subtype Classification: A Pilot Study Comparing Convolutional Neural Network Architectures	Journal of Imaging	fracture / trauma / abuse	<i>diagnosis</i> : Which CNN architecture best tells pediatric wrist fracture subtypes (buckle, greenstick, Salter-Harris) apart on radiographs? (pilot, 940 images)
4	9.8	Deep learning super-resolution of paediatric ultra-low-field MRI without paired high-field scans	bioRxiv	brain MRI access (portable low-field)	<i>reconstruction</i> : Can deep learning sharpen affordable ultra-low-field pediatric brain MRI toward high-field quality without paired high-field scans to train on?
2	9.8	Acquisition time/dose reduction in pediatric PET imaging using patch-based deep learning	EJNMMI Physics	radiation dose reduction (PET)	<i>reconstruction</i> : Can pediatric PET use one fifth of the counts with patch-based deep-learning denoising?

Radiology AI works per year, by conference



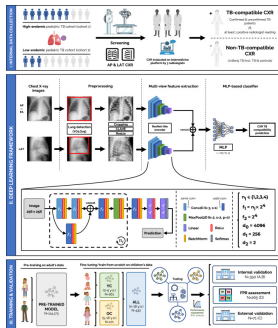
AI fracture detection in a real pediatric emergency department



Ziegner et al., Eur Radiol 2025, figure 4: pediatric fractures the software found, with its per-region sensitivity and specificity

source: cdn.ncbi.nlm.nih.gov

- **Problem:** fractures in children are missed most often by the least experienced readers, at night, in the emergency department.
- **Data:** 1,672 consecutive radiographs of children under 18 from one tertiary pediatric emergency department, plus a selected medicolegal set (proximal tibia, radial condyle); three pediatric residents read each before and after seeing the AI.
- **Method:** a commercially available, CE-marked deep-learning fracture detector (adult-derived, pediatric extension) run stand-alone and as a second reader.
- **Result:** stand-alone sensitivity 92%, specificity 83%; 100% for proximal tibia but 68% for radial condyle fractures; residents' accuracy rose from 88% to 90%, and in 2% of cases they dropped a correct diagnosis after seeing the AI.
- **Why it matters:** the first real-life pediatric ED evaluation: the gain from a cleared adult tool is real but modest, and the elbow remains the weak spot.



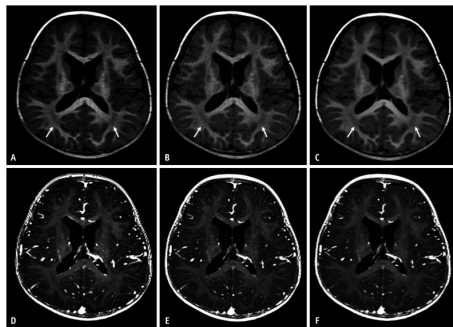
- **Problem:** pediatric pulmonary TB has vaguer symptoms and subtler radiographs than adult TB; most affected children in high-burden settings are never diagnosed.
- **Data:** pre-trained on 114,173 adult chest radiographs, then fine-tuned and tested on 918 radiographs from three pediatric TB cohorts, with lateral views.
- **Method:** a multi-view network that scores frontal and lateral films for TB-compatible findings; separate models for under-5 and 5-18 years.
- **Result:** AUC 0.90 on internal and 0.68 on external testing; lateral views helped most in the youngest children; age-specific models matched larger undifferentiated ones.
- **Why it matters:** shows both the value of adult pre-training for a rare pediatric target and the internal-to-external drop that every pediatric deployment should expect.

Capellan-Martin et al., Nat Commun 2025, figure 1: the multi-view

pTBLightNet pipeline

source: cdn.ncbi.nlm.nih.gov

Deep-learning reconstruction shortens pediatric brain MRI



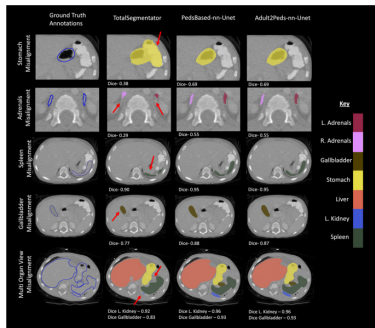
Yoo et al., Korean J Radiol 2025, figure 2: conventional (A, D) vs accelerated 3D T1 pediatric brain MRI without (B, E) and with (C, F)

deep-learning reconstruction

source: cdn.ncbi.nlm.nih.gov

- **Problem:** long 3D T1 sequences are where children move, sedation is prescribed, and scanner time is lost.
- **Data:** 46 children scanned with both the conventional and an accelerated 3D T1 protocol (pre- and post-contrast) on one 3T scanner.
- **Method:** vendor deep-learning reconstruction applied to the accelerated acquisition; noise, contrast, signal-to-noise and radiologist ratings compared with the conventional scan.
- **Result:** acquisition time cut by 29% (pre-contrast) and 41% (post-contrast); the deep-learning images had lower noise, higher SNR and CNR, better overall quality and fewer artifacts than the conventional scan.
- **Why it matters:** reconstruction AI is on the scanner today and gives children a shorter, better exam without any diagnostic model in the loop.

Children are not small adults: an adult CT segmentation model in children



Chatterjee et al., J Imaging Inform Med 2025, figure 7: pediatric CT

organ labels, ground truth vs adult-trained TotalSegmentator vs

pediatric-trained models

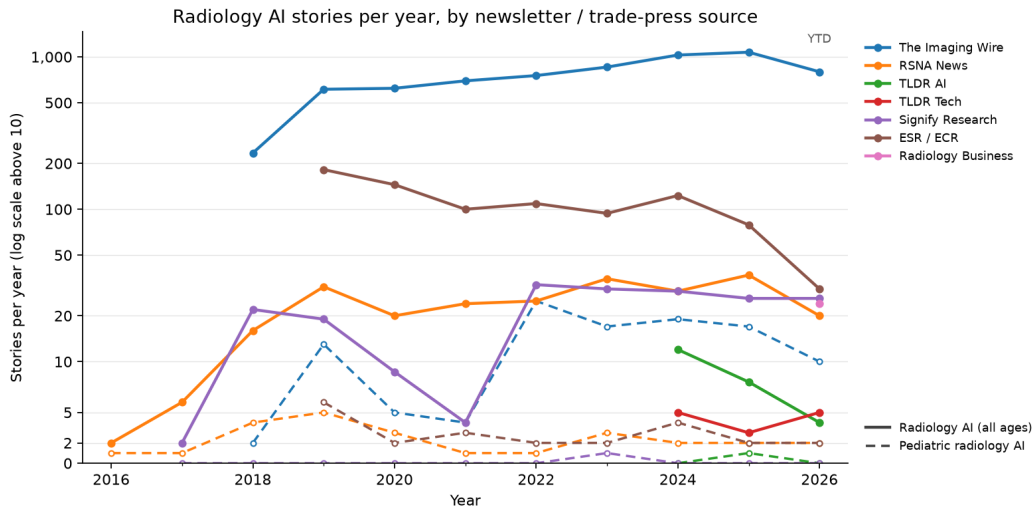
source: cdn.ncbi.nlm.nih.gov

- **Problem:** the most-used open CT organ segmentation model (TotalSegmentator) was trained on adults; nobody had measured how it behaves in children.
- **Data:** external adult ($n = 300$) and pediatric ($n = 359$) abdominal CT sets with expert organ labels.
- **Method:** TotalSegmentator scored on both sets; then a pediatric-only nnU-Net and an adult model fine-tuned on pediatric cases.
- **Result:** mean Dice fell from 0.81 in adults to 0.73 in children, worst for the adrenals (0.69 to 0.41 and 0.35), duodenum and pancreas; pediatric training or fine-tuning recovered the loss.
- **Why it matters:** the clearest quantitative case for local pediatric validation and fine-tuning of adult-derived tools before clinical use.

The biggest public pediatric radiology AI datasets

Dataset	Year	Modality	Size	Ages	Access
CBTN clinical MRI (Children's Brain Tumor Network)	2023	brain / spine tumor MRI	23,101 exams (1,526 patients); tumor masks for 370	pediatric	application (NCI CCDI)
GRAZPEDWRI-DX	2022	wrist radiographs (trauma)	20,327 images (6,091 patients)	0.2–19 y	open
RSNA Pediatric Bone Age Challenge	2017	hand radiographs	14,236 images (bone age + sex)	mean 10.6 y	open (RSNA terms)
FETAL_PLANES_DB	2020	fetal ultrasound (standard planes)	12,400 images (1,792 pregnancies)	fetal	open
ABCD Study	2018–	brain MRI, longitudinal	11,878 participants, up to 7 follow-ups	9–10 y at baseline	application (NBDC)
PediCXR / VinDr-PCXR	2023	chest radiographs	9,125 studies; 36 findings, 15 diagnoses	< 10 y	credentialed (PhysioNet)
Guangzhou pediatric chest X-ray (Kermany)	2018	chest radiographs	5,856 images (normal / bacterial / viral pneumonia)	1–5 y	open
ABIDE I + II (autism)	2012 / 2016	brain MRI (structural, resting fMRI)	2,156 subjects	5–64 y (mostly children)	open
Regensburg Pediatric Appendicitis	2023	abdominal ultrasound + labs	1,709 images (579 patients)	0–18 y	open
dHCP (Developing Human Connectome Project)	2022–24	neonatal + fetal brain MRI	886 neonatal + 297 fetal datasets, with segmentations	20–45 weeks	open (NDA)
Pediatric-CT-SEG (TCIA)	2022	chest / abdomen / pelvis CT	718 series (359 patients); 29 organ contours	5 days – 16 y	open
BraTS-PEDs (pediatric brain tumor challenge)	2023–25	brain tumor MRI	464 patients (2024 challenge); 1,649 series on TCIA	pediatric	registration / TCIA
FeTA (fetal brain tissue annotation)	2021–24	fetal brain MRI	280 volumes (120 train + 160 test); 7 tissue labels	18–35 weeks	registration

Radiology AI newsletter stories per year, by source



Pediatric radiology AI in the trade press, 2026 (year to date) (14 stories)

Date	Newsletter	Story	Paper
2026-01-07	The Imaging Wire	BrightHeart Raises EUR 11M	
2026-01-28	The Imaging Wire	AHA Publishes New Stroke Guidelines	Prabhakaran et al., 2026 (Stroke)
2026-01-28	ESR / ECR	Guerbet: Innovation and French Cooperation in the Spotlight at ECR 2026	
2026-02-01	The Imaging Wire	DL-Based CT Reduces Radiation Dose	
2026-02-08	The Imaging Wire	GE Bolsters Ultrasound with Fetal AI	
2026-02-26	ESR / ECR	MRI Unveiling at the European Congress of Radiology (ECR) 2026	
2026-03-29	The Imaging Wire	Ultrasound AI Gets Breakthrough Nod	
2026-05-10	The Imaging Wire	Pediatric MRI Safety, RTs and Remote Radiology, and PSMA-PET	
2026-06-22	RSNA News	Researchers Use AI to Establish Body Charts for CT Imaging Across the Adult Lifespan	
2026-06-28	The Imaging Wire	BrightHeart's Bright Future	Zelop et al., 2025 (Obstetrics & Gynecology)
2026-07-12	The Imaging Wire	AI's Rise in Pediatric Imaging	Shah et al., 2026 (European Radiology)
2026-08-16	The Imaging Wire	CT vs. MRI for Pediatric TBI, Helium Vulnerability, and Ditching the Disk	
2026-08-25	RSNA News	AI Challenges Fuel Innovation Beyond the Leaderboard	
2026-09-23	The Imaging Wire	Repeat MRI Scans Take Operational Toll	

Story 3: what pediatric departments actually report using

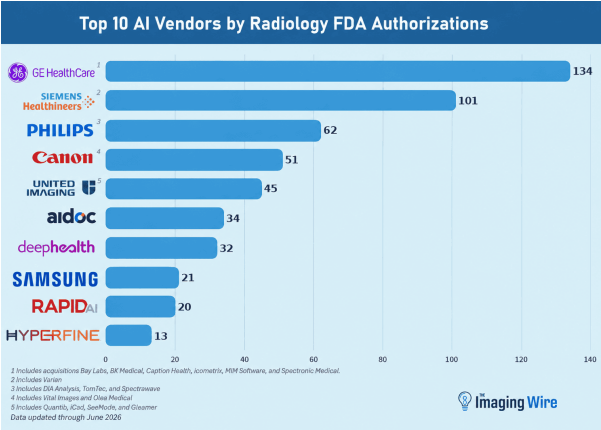
The Imaging Wire, 12 July 2026: “AI’s Rise in Pediatric Imaging” — Shah et al., *European Radiology* 2026 (doi 10.1007/s00330-026-12728-9)

- **Why it was news:** after a decade of radiology AI research, almost no data existed on what pediatric departments have actually deployed. This international survey of 18 academic pediatric radiology centers (69% response) is the first real answer.
- **Adoption is wide but shallow:** **88.9%** run at least one AI tool, yet the tools cluster on two tasks — bone age (44.4%) and segmentation / quantification (22.2%) — and only **16.7%** called AI “transformational”.
- **The barriers named:** lack of pediatric datasets (83.3%), integration into PACS and workflow (66.7%), cost (50%).
- **It matches this deck’s own data:** bone age is also the most common pediatric clinical problem in the published literature and the one task with a mature commercial product.

Take-home: the constraint is not model accuracy. It is pediatric training data and integration — both of which a children’s hospital, not a vendor, is positioned to fix.

Radiology keeps its lead in FDA AI authorizations

The Imaging Wire, 14 September 2026: FDA AI-enabled device list, updated through June 2026

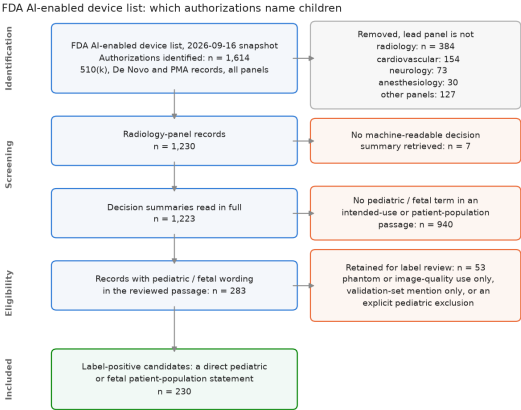


- **1,614** AI-enabled devices authorized since the FDA began tracking, up 5.9% from Q1 2026.
- **1,230** of them (76%) are radiology, the same radiology-panel total this deck's own FDA pull finds.
- Q2 2026: 89 authorizations (92 in Q1), 66 of them (74%) radiology.
- Scanner makers lead: GE HealthCare 134, Siemens Healthineers 101, Philips 62, Canon 51, United Imaging 45.
- The largest software-only vendors are Aidoc (34), DeepHealth (32) and RapidAI (20).
- The FDA is still deciding how to regulate generative AI used in direct clinical care.

Top 10 vendors by radiology FDA authorizations, counts include acquisitions

source: theimagingwire.com

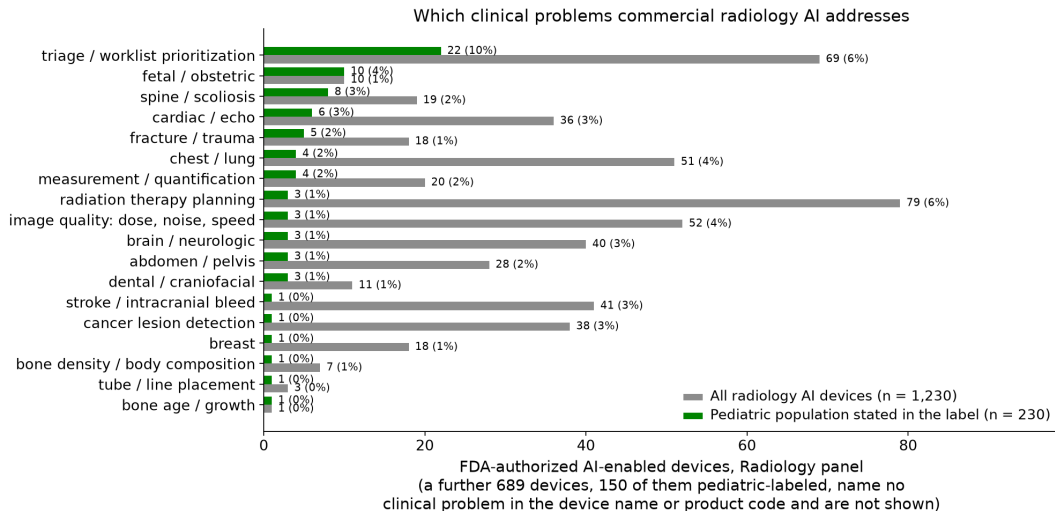
How many cleared devices actually name children?



Keyword screen of the linked FDA decision summaries, not a determination of pediatric authorization: confirm the current labeling before purchase. One row per submission, so versions and resubmissions count separately.

230 of the 1,230 radiology-panel authorizations (19%) name a pediatric or fetal population in the labeling passage reviewed; 53 more name children only as a phantom, a validation set or an exclusion, and are kept for label review.

Which clinical problems the cleared products address



Products a pediatric radiologist should know (part 1)

Product	Vendor	Clinical problem	What it does	Pediatric scope	Where it stands
BoneXpert	Visiana	bone age / growth	Automated Greulich–Pyle and Tanner–Whitehouse bone age plus a Bone Health Index, from a hand radiograph	Built for children; CE-marked and in routine European use; US research use only	Widely deployed (Europe)
BoneView	Gleamer	fracture / trauma	Fracture detection on trauma radiographs, presented as regions of interest alongside the image	US indication covers ages 2–21 for specified extremity views; pelvis, hip, femur, ribs and spine are adult-only [K222176, 2023]	Widely deployed
Rayvolve (AZtrauma)	AZmed	fracture / trauma	Fracture and MSK finding detection on radiographs; sibling modules cover chest findings, measurements and bone age	US fracture indication from age 2; the bone-age module is CE-marked only [K240845, 2024]	Widely deployed
TechCare Trauma	Milvue	fracture / trauma	Trauma radiograph triage: fractures, effusions and dislocations flagged for the reader	The decision summary states a pediatric population [K242171, 2025]	Early US deployments
EFAI Bonesuite XR Bone Age Pro	Ever Fortune.AI	bone age / growth	Greulich–Pyle bone age from a PA left hand and wrist radiograph, as an adjunct to the radiologist	FDA-cleared for ages 2–16 — the pediatric-specific US bone-age clearance [K234042, 2024]	FDA-cleared, early adoption
Physis	16 Bit	bone age / growth	Bone-age estimation with Greulich–Pyle atlas matching and draft report text	Pediatric tool; Health Canada licensed, no FDA authorization identified	Promising, limited US access
Fetal EchoScan	BrightHeart	fetal / obstetric	Flags eight morphological findings suspicious for congenital heart disease on second-trimester views	Prenatal by definition: second-trimester exams, maternal age 18+ [K252294, 2025]	Early deployments / pilots
Sonio Detect / Sonio Suspect	Sonio	fetal / obstetric	Checks that the required fetal views were obtained and are of usable quality (Detect); flags images suspicious for anomalies (Suspect)	Prenatal by definition; every FDA entry states a fetal population [K252433, 2026]	Early deployments / pilots
Delivery Date AI	Ultrasound AI	fetal / obstetric	Predicts the delivery date from a second- or third-trimester ultrasound when dating is unreliable	Prenatal; the first De Novo of its kind (a new device class) [DEN250007, 2026]	Newly authorized, unproven in practice
Swoop Portable MR + BrainInsight	Hyperline	brain / neurologic	Bedside low-field brain MRI whose usable image quality comes from deep-learning reconstruction; BrainInsight adds automated volumes and midline shift	Decision summaries state a pediatric population; the natural use is the NICU and PICU, where transport is the risk [K253489, 2025]	In pediatric pilots
spineEOS / VEA Align	EOS imaging (Alphatec)	spine / scoliosis	Automated 3D spine and lower-limb modeling from biplanar low-dose full-body radiographs	Scoliosis follow-up is repeated imaging of a growing child, so the dose saving compounds [K251747, 2025]	Widely deployed (pediatric orthopedics)
CoLumbo	Smart Soft Healthcare	spine / scoliosis	Spine MRI segmentation, measurement and structured reporting, lumbar and cervical	Decision summaries state a pediatric population; no pediatric-specific performance reported [K254015, 2026]	Early deployments

Square brackets give the FDA submission supporting the pediatric claim and its decision year; a blank means no US authorization was identified for that product. “Where it stands” is our reading of deployment, not a measured adoption rate. A company’s total of FDA AI entries is not a pediatric credential — the large vendors’ entries are mostly whole scanners whose labels list pediatric imaging as one clinical application (see the authorizations-per-company figure).

Products a pediatric radiologist should know (part 2)

Product	Vendor	Clinical problem	What it does	Pediatric scope	Where it stands
BriefCase-Triage (CARE modules)	Aidoc	triage / worklist prioritization	Worklist triage on CT and CTA: intracranial hemorrhage, pulmonary embolism, pneumothorax, cervical-spine and rib fractures, free gas and more	Most entries state a pediatric population; the evidence behind them is adult, so treat pediatric use as local validation work [K253578, 2026]	Widely deployed
Viz LVO / Viz ICH	Viz.ai	stroke / intracranial bleed	Detects large-vessel occlusion and intracranial hemorrhage and pushes the case to the stroke team's phones	No pediatric population statement found in the screened summaries; pediatric stroke use would be off-label	Widely deployed (adult)
qXR-TB	Qure.ai	chest / lung	Chest radiograph screening for tuberculosis, built for high-burden settings with few readers	Vendor reports CE-marked use from age 0–15; US eligibility varies by module	Deployed in global-health screening
AIR Recon DL / TrueFidelity	GE HealthCare	image quality: dose, noise, speed	Deep-learning reconstruction: shorter MRI acquisitions at the same quality, and CT images at lower dose	Sold across ages; the pediatric benefit is fewer sedations and less dose, and it is the AI most children actually meet	Widely deployed
Deep Resolve	Siemens Healthineers	image quality: dose, noise, speed	MRI deep-learning denoising and resolution enhancement that supports accelerated acquisition	Vendor publishes pediatric protocols; scan-time reduction is the pediatric argument	Widely deployed
AIce	Canon Medical	image quality: dose, noise, speed	Deep-learning reconstruction for both CT (noise, dose) and MRI (noise, resolution, scan time)	Vendor publishes pediatric CT galleries; protocol and version determine what is available	Widely deployed
SmartSpeed / Precise Image	Philips	image quality: dose, noise, speed	MRI acceleration with deep-learning reconstruction (SmartSpeed); CT reconstruction at reduced dose (Precise Image)	Sold across ages; pediatric protocols are the vendor's own, not a separate indication	Widely deployed
SubtleMR / SwiftMR	Subtle Medical / AIRS Medical	image quality: dose, noise, speed	Vendor-neutral MRI post-processing: denoising and sharpening so a shorter acquisition still reads	No adult-only restriction in the reviewed US indications; SwiftMR's 2026 validation includes ages 0–21	Deployed, vendor-neutral retrofit
Butterfly iQ3; Clarius Bladder AI	Butterfly Network / Clarius	point-of-care ultrasound	Handheld ultrasound with on-device AI: bladder volume, gestational age, ejection fraction, lung and nerve tools	The scanners are labeled for adult and pediatric imaging, but each AI tool carries its own age scope and most are adult-only; Clarius Bladder AI covers ages 2 and over [K232257, 2023]	Scanners widely deployed; AI tools mostly adult-scoped
Arterys Cardio DL	Arterys (Tempus)	cardiac / echo	Cardiac MRI segmentation, flow and volume quantification with a cloud workflow	One of the earliest AI clearances to name neonates, infants, children and adolescents outright (2017) [K163253, 2017]	Historically important; now a Tempus product
Acorn 3D	Mighty Oak Medical	spine / scoliosis	Automated vertebral segmentation from CT to build patient-specific 3D models and surgical guides	Decision summaries state a pediatric population; the use case is pediatric deformity surgery [K260322, 2026]	Niche, surgical
Second Opinion Pediatric	Pearl	dental / craniofacial	Detects caries and other findings on pediatric dental radiographs	The rare device whose name and indication are pediatric from the start [K243893, 2025]	Deployed in dentistry

Square brackets give the FDA submission supporting the pediatric claim and its decision year; a blank means no US authorization was identified for that product. “Where it stands” is our reading of deployment, not a measured adoption rate. A company's total of FDA AI entries is not a pediatric credential — the large vendors' entries are mostly whole scanners whose labels list pediatric imaging as one clinical application (see the authorizations-per-company figure).

Product 1: BoneXpert — the most-validated pediatric AI you cannot buy

Visiana, Denmark (bonexpert.com) — bone age / growth

- **What it does:** automated Greulich–Pyle and Tanner–Whitehouse bone age from a single hand radiograph, plus a **Bone Health Index** from cortical thickness and an adult-height predictor. Not a second reader — it replaces the manual rating.
- **Why it is the benchmark:** designed for children from the start, in routine European clinical use since 2009, and the comparator nearly every academic bone-age paper measures itself against. It is the one pediatric AI with a decade of unglamorous production experience behind it.
- **The catch:** CE-marked in Europe; in the United States the vendor labels it **research use only**. There is no FDA authorization. The only pediatric-specific US bone-age clearance is a different product, EFAI Bonesuite XR Bone Age Pro (ages 2–16, K234042, 2024).
- **What it tells us about the market:** pediatric-native tools exist and work. The bottleneck is a US regulatory and commercial path, not the science.

Take-home: when someone says “there is no pediatric AI”, the accurate statement is that the best pediatric AI is not licensed here.

Product 2: BoneView — read the anatomy, not just the age range

Gleamer, France — fracture / trauma; US clearance K222176 (BoneView 1.1-US, 2023)

- **What it does:** flags suspected fractures on trauma radiographs as regions of interest beside the image. The international suite also covers dislocations, effusions and focal bone lesions; **the US clearance is fracture detection only.**
- **The indication is narrower than “ages 2–21”:** K222176 permits ages 2–21 for **specified extremity views**, clavicle included. Pelvis, hip, femur, ribs and spine are **adult-only** in that summary — a child with a suspected hip or rib fracture is outside the pediatric indication.
- **What the real-life pediatric evidence shows:** the one pediatric ED evaluation of a cleared fracture detector (Ziegner et al., *Eur Radiol* 2025, 1,672 radiographs) found 92% stand-alone sensitivity but only **68% for radial condyle fractures**, and residents' accuracy rose 88% → 90%.
- **The failure mode to watch:** in 2% of cases readers abandoned a correct diagnosis after seeing the AI.

Take-home: the most mature pediatric AI market still needs local validation by anatomy, and a reader policy for disagreement.

Product 3: Aidoc BriefCase — a pediatric label is not pediatric evidence

Aidoc, Israel — CT/CTA triage and worklist prioritization; e.g. K253578 (2026)

- **What it does:** reads CT and CTA as they arrive and reorders the worklist — intracranial hemorrhage, pulmonary embolism, pneumothorax, cervical-spine and rib fractures, free gas. Triage, not diagnosis, is the product.
- **The number that matters here:** Aidoc holds **23 of the 230 pediatric-label-positive** radiology authorizations in our FDA screen, more than any other company.
- **What that does and does not mean:** the decision summaries state a pediatric patient population, but the **performance data behind them is adult**. A population statement in an intended-use paragraph is a candidate for label review, not evidence of testing in children.
- **Why it matters operationally:** triage silently changes the order a pediatric radiologist sees cases in. A model tuned on adult prevalence will mis-prioritize a pediatric list, and nobody sees the ranking it produced.

Take-home: “FDA-cleared, pediatric population included” is this market’s most common and most misleading claim. Ask for pediatric subgroup performance; usually there is none.

Product 4: deep-learning reconstruction — the AI children actually meet

GE AIR Recon DL / TrueFidelity; Siemens Deep Resolve; Canon AiCE; Philips SmartSpeed / Precise Image

- **What it does:** reconstructs MRI and CT from noisier, faster or lower-dose raw data — shorter MRI at comparable quality, CT at reduced dose. It is not interpretive: it makes the picture the radiologist then reads.
- **Why it is the biggest pediatric win on this list:** a shorter MRI is **fewer sedations**; a lower-dose CT is **less radiation to a growing child**. Neither requires anyone to trust a diagnostic model.
- **The regulatory shape is different:** these are sold across ages as part of the scanner, with vendor-published pediatric protocols rather than a pediatric indication. Philips holds 28 pediatric-label-positive entries, **one** of which is software; the rest are scanner labels listing pediatric imaging as an application.
- **The caveat:** dose and time reductions are not fixed properties of the product name — they depend on task, patient size and configuration, and must be measured locally.

Take-home: most of the pediatric benefit already installed here is non-interpretive AI nobody calls AI, and it is where a validation program should start.

Kamran review: broader research, stronger validation

Kamran et al., *Pediatric Radiology* 2026 (doi 10.1007/s00247-025-06462-5)

- **A large but concentrated literature:** 789 pediatric-focused original articles (2005–August 2024); 91% addressed image interpretation/diagnosis. Leading applications were bone age (88), pneumonia (65), and brain lesions/tumors (54).
- **Generalizability needs stronger evidence:** 65% relied on local hospital datasets. The authors call for diverse, multicenter datasets and prospective validation across pediatric ages, sites, and disease-prevalence settings.
- **Broaden the research agenda:** communication, education, policy, and stakeholder perspectives were underrepresented. Include children, families, and clinical teams in development and evaluation.

Take-home: prioritize representative data, validation, and neglected clinical needs. This review maps research activity; it does not compare model accuracy or establish clinical readiness.

ACR white paper: the pediatric AI gap

Sammer et al., *J Am Coll Radiol* 2023;20:730–737 (doi 10.1016/j.jacr.2023.06.003)

- **Scope:** ACR Pediatric AI Workgroup white paper framing limited access to pediatric AI as a health equity issue.
- **Availability:** at the time of the review, only 6 of more than 200 FDA-cleared tools in the ACR catalog indicated pediatric use (about 3%).
- **Why adult AI may fail:** growth, disease patterns, and imaging protocols differ; published studies show uneven performance across pediatric subgroups.
- **Barriers:** scarce labeled pediatric data, consent and privacy concerns, and weak financial incentives for smaller pediatric markets.
- **Recommendations:** validate in children, disclose age-specific evidence, fund pediatric development, involve families, and protect pediatric workflows. The authors introduce *Image IntelliGently* to advance this work.

Multisociety statement: safe pediatric AI implementation

Shelmerdine et al., *J Am Coll Radiol* 2026;23:89–101 (doi 10.1016/j.jacr.2025.08.019)

ACR, ESPR, SPR, SLARP, AOSPR, and SPIN propose four pillars for safe adoption:

- **Regulation and purchasing:** pediatric labeling and proposed safety ratings; diverse datasets, subgroup performance, transparency, and cybersecurity.
- **Implementation and integration:** define a local clinical need, select tools validated in children, pilot before scaling, and involve pediatric specialists and families.
- **Interpretation and postmarket surveillance:** retain human oversight; audit performance and patient outcomes; manage discrepancies, report adverse events, and address automation bias.
- **Education:** foundational AI literacy for all staff, pediatric-specific training, continuing education, and patient/public engagement.

Worth knowing as a radiologist

- **Deep Learning: A Primer for Radiologists** (Radiographics paper, 2017): a helpful introduction to the principles and applications of deep learning in radiology; essential for understanding how AI tools work and their limitations
- **RSNA Pediatric Bone Age Challenge (2017)** (benchmark / dataset): a landmark pediatric benchmark; useful for understanding automated bone-age accuracy and its limits
- **Gleamer BoneView / AZmed Rayvolve** (clinical products): examples of fracture-detection assistance with pediatric indications; eligibility depends on age, anatomy, and product version
- **FDA AI-enabled device list and linked labeling** (regulatory resource): a starting point for checking authorized use and pediatric age ranges; the list is not exhaustive
- **CLAIM checklist (2024 update)** (reporting guideline): check whether data, reference standards, and testing are adequately described; reporting completeness alone does not establish clinical benefit
- **TotalSegmentator** (research tool): automated CT / MR organ segmentation for volumes and body composition; check accuracy in children before trusting measurements
- **nnU-Net** (research framework): a strong baseline for training segmentation models; ask whether a new method improves on an appropriate comparator
- **MedSAM / segment-anything models** (research models): prompt-guided contours can assist annotation; radiologists still need to review and correct the output
- **MONAI** (research framework): infrastructure for imaging-AI development; useful when collaborating with a research team, rather than a ready-to-use diagnostic tool
- **NVIDIA NV-Reason-CT** (open research model, Sept 2026): a 3D CT vision-language model that drafts chest / abdominal CT findings with step-by-step reasoning and answers questions; open weights, research use only, not FDA-cleared, and no pediatric data or age range reported

Explore our database: <https://pedrad-ai.vercel.app>

Pediatric Radiology AI Database



Articles, open source software, trade-press stories, commercial devices and public datasets for AI in pediatric radiology. Search, filter columns by value, sort, pick columns, and export any view as CSV.

Articles 3,496

Open source software 339

Newsletters 161

Commercial 997

Datasets 13

The FDA's list of AI-enabled medical devices, Radiology panel only (snapshot 2026-09-17; [source](#)), one row per device: a device the FDA authorized more than once (a new version or indication) is shown at its most recent authorization, with the earlier ones listed, so the 1,230 authorizations make 994 rows. The FDA describes the list as noncomprehensive. The 3 rows marked **not FDA-authorized** are pediatric products on our curated list that have no US authorization (for example BoneXpert, CE-marked and US research use only). **Pediatric use** is yes for the 172 authorized devices with at least one authorization whose linked label states a pediatric or fetal patient population or intended use (2026-09-16 screen); hover the yes for which. About half are whole scanners whose labels list pediatric imaging as one clinical application, so confirm the current authorization before purchase. **Curated pediatric product** marks the 23 products with a documented pediatric indication reviewed by hand on 2026-09-16; on an FDA row it is a company-level flag, not a claim about that row's authorization.

Search

Clear

Columns ▾

Export CSV

Examples: `pediatric:yes` `company:="GE HealthCare"` `date>=2025` `fetal OR pediatric` `us_status:="not FDA-authorized"` [show fields](#)

12 of 997 rows match.

Device	Company ▾	Decision date	Submission	US status ▾	Earlier authorizations	Product code ▾	Pediatric-use screen ▾	Pediatric use ▴ ▾
▶ EFAI Chestsuite XR Malpositioned ETT Assessment System (ETT-XR-100)	Ever FortuneAI	2025-02-20	K242821	FDA-authorized	—	QAS	label-positive candidate	yes
▶ EFAI Bonesuite XR Bone Age Pro Assessment System (BAP-XR-100)	Ever FortuneAI	2024-06-07	K234042	FDA-authorized	—	QIH	label-positive candidate	yes
▶ EFAI CARDIOSUITE CTA ACUTE AORTIC SYNDROME ASSESSMENT SYSTEM	Ever FortuneAI	2024-04-08	K240291	FDA-authorized	—	QAS	label-positive candidate	yes
▶ EFAI ERSUITE CT APPENDICITIS ASSESSMENT SYSTEM (APPEN-CT-100)	Ever FortuneAI	2026-04-22	K253163	FDA-authorized	—	QAS	—	no
▶ Seg Pro V3 (RT-300)	Ever FortuneAI	2026-01-28	K251306	FDA-authorized	—	QKB	—	no
▶ EFAI Neurosuite CT Midline Shift Assessment System (MLS-CT-100)	Ever FortuneAI	2024-12-06	K241923	FDA-authorized	—	QAS	—	no
▶ EFAI NeuroSuite CT ICH Assessment System	Ever FortuneAI	2023-10-04	K231025	FDA-authorized	—	QAS	—	no
▶ EFAI RTSUITE CT HCAP-Segmentation System	Ever FortuneAI	2023-09-25	K231928	FDA-authorized	—	QKB	—	no
▶ EFAI ChestSuite XR Pneumothorax Assessment	Ever Fortune AI	2022-11-08	K221552	FDA-authorized	—	QFM	—	no

Recommendations for pediatric radiologists

- AI development is constrained by scarce public datasets, fragmented institutional data, rare diseases, and heterogeneous reporting practices.
- Choose one task worth improving in your practice: bone-age measurements, fracture detection, or reducing CT dose / MRI scan time.
- Before adopting a tool, ask which children it was tested on. Check the intended age range and test it on your own cases, including younger children and unusual anatomy.
- In a local pilot, measure what changes with AI: reading time, missed findings, false alarms, dose, or scan time. Review errors and keep checking after rollout.
- Help build the evidence we need: choose a pediatric problem you see often, label cases carefully, and work with other hospitals to test whether a model works beyond the site that built it.